

GitLab Merge Requests API Specification

Complete Endpoint Reference for Custom Git Client Implementations (REST API v4)

Client Connection & Authentication Headers

Base URL: `https://<gitlab-host>/api/v4` • Authorization: `PRIVATE-TOKEN: <token>` or `Authorization: Bearer <token>`.

Path parameter `:id` accepts numeric Project ID (e.g., `105`) or URL-encoded path (e.g., `group%2Fproject`). `:mr_iid` is the MR number.

1. MR LIFECYCLE & BRANCH OPERATIONS

Core Resource

METHOD	ENDPOINT	DESCRIPTION & KEY PARAMETERS
GET	/projects/:id/merge_requests	List MRs in project (state, labels, author, milestone).
GET	/merge_requests	Global list of accessible MRs across all projects.
GET	/projects/:id/merge_requests/:mr_iid	Get detailed metadata, branch states, and merge status.
POST	/projects/:id/merge_requests	Create MR (source_branch, target_branch, title).
PUT	/projects/:id/merge_requests/:mr_iid	Update title, description, target branch, or state_event.
DELETE	/projects/:id/merge_requests/:mr_iid	Delete an MR permanently (requires Maintainer+).
PUT	/projects/:id/merge_requests/:mr_iid/merge	Accept/Merge MR (squash, remove_source_branch, sha).
PUT	/projects/:id/merge_requests/:mr_iid/rebase	Trigger rebase of source branch against target branch.
PUT	/projects/:id/merge_requests/:mr_iid/cancel_merge_when_pipeline_succeeds	Cancel auto-merge queued on pipeline completion.
PUT	/projects/:id/merge_requests/:mr_iid/reset_approvals	Reset approval count and clear assigned sign-offs.

2. DIFFS, CHANGES & COMMIT HISTORY

Code Review Engine

METHOD	ENDPOINT	DESCRIPTION & KEY PARAMETERS
GET	/projects/:id/merge_requests/:mr_iid/diffs	Get file changes, unified diff hunks, and line offsets.
GET	/projects/:id/merge_requests/:mr_iid/raw_diff	Stream unified patch diff text directly.
GET	/projects/:id/merge_requests/:mr_iid/commits	List commit objects included in the MR branch.
GET	/projects/:id/merge_requests/:mr_iid/versions	List diff versions (contains base_sha, start_sha, head_sha).
GET	/projects/:id/merge_requests/:mr_iid/versions/:version_id	Get specific diff version compare snapshot.
GET	/projects/:id/merge_requests/:mr_iid/context_commits	List cherry-picked context commits included for review.
POST	/projects/:id/merge_requests/:mr_iid/context_commits	Inject external context commits into review scope.
DELETE	/projects/:id/merge_requests/:mr_iid/context_commits	Remove context commits from review view.

3. THREADED DISCUSSIONS (LINE COMMENTS & RESOLVING)

Interactive Reviews

METHOD	ENDPOINT	DESCRIPTION & PARAMETERS
GET	/projects/:id/merge_requests/:mr_iid/discussions	List all threads (line diffs and general threads).
POST	/projects/:id/merge_requests/:mr_iid/discussions	Start thread (body, position object for code lines).
GET	/projects/:id/merge_requests/:mr_iid/discussion/:disc_id	Get single thread object with note tree.
PUT	/projects/:id/merge_requests/:mr_iid/discussion/:disc_id	Resolve or unresolve entire thread (resolved: true/false).
POST	/projects/:id/merge_requests/:mr_iid/discussion/:disc_id/notes	Reply to thread with new message body.
GET	/projects/:id/merge_requests/:mr_iid/discussion/:disc_id/notes/:note_id	Get individual comment within a thread.
PUT	/projects/:id/merge_requests/:mr_iid/discussion/:disc_id/notes/:note_id	Edit comment text or resolve individual note.
DELETE	/projects/:id/merge_requests/:mr_iid/discussion/:disc_id/notes/:note_id	Delete individual comment from discussion.

4. DRAFT NOTES & BATCH REVIEW SUBMISSION

Pending Reviews

METHOD	ENDPOINT	DESCRIPTION & PARAMETERS
GET	/projects/:id/merge_requests/:mr_iid/draft_notes	List current user's unpublished draft notes.
POST	/projects/:id/merge_requests/:mr_iid/draft_notes	Save pending review comment (body, position).
GET	/projects/:id/merge_requests/:mr_iid/draft_notes/:draft_id	Get specific draft review comment.
PUT	/projects/:id/merge_requests/:mr_iid/draft_notes/:draft_id	Update text or target line of draft comment.
DELETE	/projects/:id/merge_requests/:mr_iid/draft_notes/:draft_id	Discard single draft review comment.
POST	/projects/:id/merge_requests/:mr_iid/draft_notes/:draft_id/publish	Publish single pending draft comment.
POST	/projects/:id/merge_requests/:mr_iid/draft_notes/bulk_publish	Submit complete review: publish all drafts at once.

5. APPROVALS & GOVERNANCE RULES

Compliance & Sign-off

METHOD	ENDPOINT	DESCRIPTION & PARAMETERS
GET	/projects/:id/merge_requests/:mr_iid/approvals	Get approval requirements, approved_by, and approvers.
POST	/projects/:id/merge_requests/:mr_iid/approve	Approve MR (optional: sha to prevent race conditions).
POST	/projects/:id/merge_requests/:mr_iid/unapprove	Revoke approval from MR.
GET	/projects/:id/merge_requests/:mr_iid/approval_state	Evaluate rule compliance (checks which rules passed).
GET	/projects/:id/merge_requests/:mr_iid/approval_rules	List MR-specific approval rule criteria.
POST	/projects/:id/merge_requests/:mr_iid/approval_rules	Create approval rule (name, approvals_required, merge_id)

6. TOP-LEVEL COMMENTS (NOTES)

Unthreaded Timeline

METHOD	ENDPOINT	DESCRIPTION & PARAMETERS
GET	/projects/:id/merge_requests/:mr_iid/notes	List top-level unthreaded comments and system logs.
POST	/projects/:id/merge_requests/:mr_iid/notes	Post main activity comment (body text).
GET	/projects/:id/merge_requests/:mr_iid/notes/:note_id	Get single top-level note content.
PUT	/projects/:id/merge_requests/:mr_iid/notes/:note_id	Update top-level note text body.
DELETE	/projects/:id/merge_requests/:mr_iid/notes/:note_id	Delete top-level comment.

7. CI/CD PIPELINES & STATUS CHECKS

Automation Gates

METHOD	ENDPOINT	DESCRIPTION & PARAMETERS
GET	/projects/:id/merge_requests/:mr_iid/pipelines	List CI pipelines triggered by the MR branch.
POST	/projects/:id/merge_requests/:mr_iid/pipelines	Manually trigger pipeline for merge request.
GET	/projects/:id/merge_requests/:mr_iid/external_status_checks	List status of external webhook merge gates.
POST	/projects/:id/merge_requests/:mr_iid/status_check_responses	Pass/fail response for external status check.

8. EMOJI REACTIONS (AWARDS)

Feedback

METHOD	ENDPOINT	DESCRIPTION & PARAMETERS
GET	/projects/:id/merge_requests/:mr_iid/award_emoji	List emojis awarded directly to the MR.
POST	/projects/:id/merge_requests/:mr_iid/award_emoji	React to MR with emoji (name: 'thumbsup').
DELETE	/projects/:id/merge_requests/:mr_iid/award_emoji/:award_id	Remove emoji reaction from MR.
GET	/projects/:id/merge_requests/:mr_iid/notes/:note_id/award_emoji	List emoji reactions on a specific comment.
POST	/projects/:id/merge_requests/:mr_iid/notes/:note_id/award_emoji	React to a comment with an emoji.
DELETE	/projects/:id/merge_requests/:mr_iid/notes/:note_id/award_emoji/:id	Remove emoji reaction from comment.

9. ISSUE LINKS, PARTICIPANTS & TIME TRACKING

Metadata

METHOD	ENDPOINT	DESCRIPTION & PARAMETERS
GET	/projects/:id/merge_requests/:mr_iid/closes_issues	List issues resolved automatically upon merge.
GET	/projects/:id/merge_requests/:mr_iid/participants	List user accounts participating in the MR.
GET	/projects/:id/merge_requests/:mr_iid/time_tracking_stats	Get time estimate and total spent time logged.
POST	/projects/:id/merge_requests/:mr_iid/time_estimate	Set time estimate duration (e.g., '3h 30m').
POST	/projects/:id/merge_requests/:mr_iid/add_spent_time	Log spent review/coding time (e.g., '1h').